

About This Course

Our world is full of objects. To model our beautiful world, a straight-forward way is object-oriented programming. This course uses the Java programming language to introduce the principles of object-oriented programming including: classes, objects, interfaces, references, methods and fields; types of inheritance, overriding, polymorphism, overloading, exception handling, generics, recursion, design by contract, file I/O, use of standard APIs and the implementation of some basic data structures and algorithms.

Dates & Time

1. **Please refer to mosaic for detailed information of the dates and time of this course.**
2. All lectures will be in person and no video recording.
3. **Tutorials:**
 - **T01 (in-person):** Jimmy Yan
 - **T02 (online in September and in-person afterwards):** Mohamadreza Sabeghi
 - **T03 (in-person):** Vanessa Lai
 - **T04 (in-person):** Baran Shajari
4. **Office hours:**
 - Instructor: Dr. Chu (chul9@mcmaster.ca)
 - Time: 1:30PM - 2:30PM on Thursday
 - Location: ITB 213
 - TA: Mohamadreza Sabeghi (TA lead) (sabeghim@mcmaster.ca):
 - Time: 3:30PM - 4:30PM on Wednesday
 - Location: ITB 102
 - TA: Baran Shajari (shajarib@mcmaster.ca):
 - Time: pending
 - Location: pending

- TA: Vanessa Lai (laiv3@mcmaster.ca)
 - Time: 10:30AM - 11:30AM on Tuesday
 - Location: pending
- TA: Jimmy Yan (yanj94@mcmaster.ca):
 - Time: 2:30PM - 3:30PM on Wednesday
 - Location: pending

5. Exams

- **Mid-term Exam** (in-person, invigilated by SAS)
 - Time: 11:30am - 12:20pm on October 22, 2024.
 - Duration: 50 minutes.
 - Location: pending
- **Final Exam** (in-person, supervised by the registrar office):
 - Time: pending
 - Duration: 2hours and 30 minutes.
 - Location: pending

Textbook and references

1. Ralph Morelli, Ralph Walde, Beryl Hoffman, "Java, Java, Java: Object-Oriented Problem Solving" [[Link](#)]

Grading Scheme

1. Five regular assignments (3% per assignment, 15% in total)
2. One project assignment (15% in total)
3. Mid-term exam (30%)
4. Final exam (40%)
5. Bonus (5% additional, the final grade will not exceed 100%)

Important Policies:

1. Please follow exactly the above format to prepare and submit your answers. Violating the format is subject to a penalty of 20% of the marks of this assignment.
2. Not all weeks have assignments. The due time of assignments may vary according to the posted information on Avenue 2 Learn (A2L).
3. The grade of an assignment will be posted on A2L seven days after the due time. The answers will be explained in the tutorials next week.
4. If you MSAF an assignment, the due time of the assignment will be extended, however, the extended deadline will not pass the time when the grades of the assignment are posted on A2L.
5. An overdue submission is subject to a penalty of 20 marks per hour exceeding the submission deadline. The penalty is calculated by:
 $\text{penalty} = \text{floor}(\text{exceed hours}) * 20$ marks. For example, if you exceed 0.9 hours, the $\text{penalty} = \text{floor}(0.9) * 20 = 0$; if you exceed 2.1 hours, the $\text{penalty} = \text{floor}(2.1) * 20 = 40$.
6. If you do not agree on any marks of an assignment or an exam, you can appeal within seven days after the marks are published by sending the instructor an email. After seven days, the marks will be finalized, and no change is allowed.
7. The instructor reserves the right to conduct deferred examinations orally.

Teaching format

1. **How to attend this course?** Lectures, tutorials and office hours will all be in person and will not be recorded. Slides will be available on Avenue.
2. **Where to find useful materials?** We will use Avenue for most of the activities related to this course. Important announcements will be posted in the general channel of the MS team of this course.
3. **IMPORTANT Note:** All in-person activities will strictly follow all related policies of the university. Course related information will sometimes be sent to your mcmaster.ca email accounts and/or posted on the course website and/or MS Teams. It is your responsibility to check all the information channels regularly.

Course Outline

Chapter00: Computers, Objects, Java, OOP Principles and Maven

- Lecture01: A Brief Introduction to the Course [[slides](#)]
- Lecture02: The Mental Landscape of Java [[slides](#)]
- Lecture03: OOP Design Principles [[slides](#)]
- Lecture04: Introduction to Maven [[slides](#), [code](#)]

Chapter01: Java Program Design and Development

- Lecture05: Java Program Design [[slides](#)]
- Lecture06: Java Language Elements [[slides](#)]

Chapter02: Objects - Using, Creating, and Defining

- Lecture07: Using String Objects [[slides](#)]
- Lecture08: Class Definition [[slides](#)]
- Lecture09: Case Study Simulating a Two Person Game [[slides](#)]

Chapter03: Methods - Communicating with Objects

- Lecture10: Passing Information to An Object [[slides](#)]
- Lecture11: Constructors and Retrieving Information from Objects [[slides](#)]
- Lecture12: Flow of Control [[slides](#)]

Chapter04: Input/Output - Designing the User Interface

- Lecture13: CommandLine User Interface [[slides](#)]
- Lecture14: Text Input and Output [[slides](#), [code](#)]

Chapter05: Java Data and Operators

- Lecture15: Variables and Types [[slides](#), [code](#)]
- Lecture16: Java Expressions [[slides](#), [code](#)]

Chapter06: Strings and String Processing

- Lecture17: String Basics [[slides](#)]
- Lecture18: Keyword Search and Retrieving Part of String [[slides](#)]
- Lecture19: Processing and Comparing Strings [[slides](#)]

Chapter07: Inheritance and Polymorphism

- Lecture20: Programming with Objects [[slides](#), [code](#)]
- Lecture21: Inheritance and the Implementation of Polymorphism [[slides](#), [code](#)]
- Lecture22: Abstract Classes and Interfaces [[slides](#), [code](#)]

Chapter08: Exceptions - When Things Go Wrong

- Lecture23: Handling Exceptions [[slides](#), [code](#)]

Chapter09: Arrays and Array Processing

- Lecture24: Array Details and Array Processing [[slides](#), [code](#)]
- Lecture25: ArrayList Searching Sorting and Polymorphism [[slides](#), [code](#)]
- Lecture26: Two Dimensional Array and Recursion [[slides](#), [code](#)]

Chapter10: Files and Streams - Input Output Techniques

- Lecture27: I/O Streams Readers and Writers [[slides](#), [code](#)]
- Lecture28: A Deeper Look into Files [[slides](#), [code](#)]
- Lecture29: Programming with Files [[slides](#), [code](#)]

Chapter11: Generic Classes and Collection Classes

- Lecture30: Generic Programming [[slides](#), [code](#)]
- Lecture31: Lists and Sets [[slides](#), [code](#)]
- Lecture32: Map Interface [[slides](#), [code](#)]
- Lecture33: Programming with JCF [[slides](#), [code](#)]
- Lecture34: Writing Generic Classes [[slides](#), [code](#)]

Chapter12: Advanced Topics

- Lecture35: Covariance and ContraVariance [[slides](#)]
- Lecture36: Design by Contract [[slides](#)]
- Lecture37: Object Oriented Programming Patterns [[slides](#)]

Tutorial Materials

Tutorial01 [[slides](#)]

Statements

Academic Dishonesty

Students are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials earned are rooted in principles of honesty and academic integrity. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (*notation reads: "Grade of F assigned for academic dishonesty"*), and/or suspension or expulsion from the university. **It is the students' responsibility to understand what constitutes academic dishonesty.** For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <http://www.mcmaster.ca/academicintegrity>.

Please be reminded that using AI generated contents in your assignments and exams will be considered an infringement of academic dishonesty.

Authenticity and Plagiarism Detection

Some courses may use a web-based service (<https://www.turnitin.com/>) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly

to Turnitin.com or via an online learning platform (e.g. Avenue 2 Learn, etc.) using plagiarism detection (a service supported by <https://www.turnitin.com/>) so it can be checked for academic dishonesty. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to <https://www.mcmaster.ca/academicintegrity/>.

Courses with An Online Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online. It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation

Students who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s Academic Accommodation of Students with Disabilities policy. Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office *normally within 10 working days* of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Requests for Relief of Missed Academic Term Work

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar “Requests for Relief for Missed Academic Term Work”.

Normally, a relief is in the form of an extended deadline which does not exceed the time when formal answers to the academic work (i.e., assignments, projects, exams, etc) are released. The instructor reserves the right to define what is a proper relief.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.

